



ACROPOLIS
Enlightening wisdom



Presentation on Universal AI Autofiller

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Use Case Diagram

- **Use Case Diagram – Smart AI Autofill System**

1. **Definition:**

Shows all actors and how they interact with the AI-based auto-fill system that detects, processes, and fills forms automatically across applications.

2. **Actors & Roles:**

User Creates and manages profiles, scans documents, sets preferences, and uses the auto-fill functionality.

Android System (Accessibility Services / Autofill Framework):

Provides access to screen content, enables detection of input fields, and supports automatic form filling.

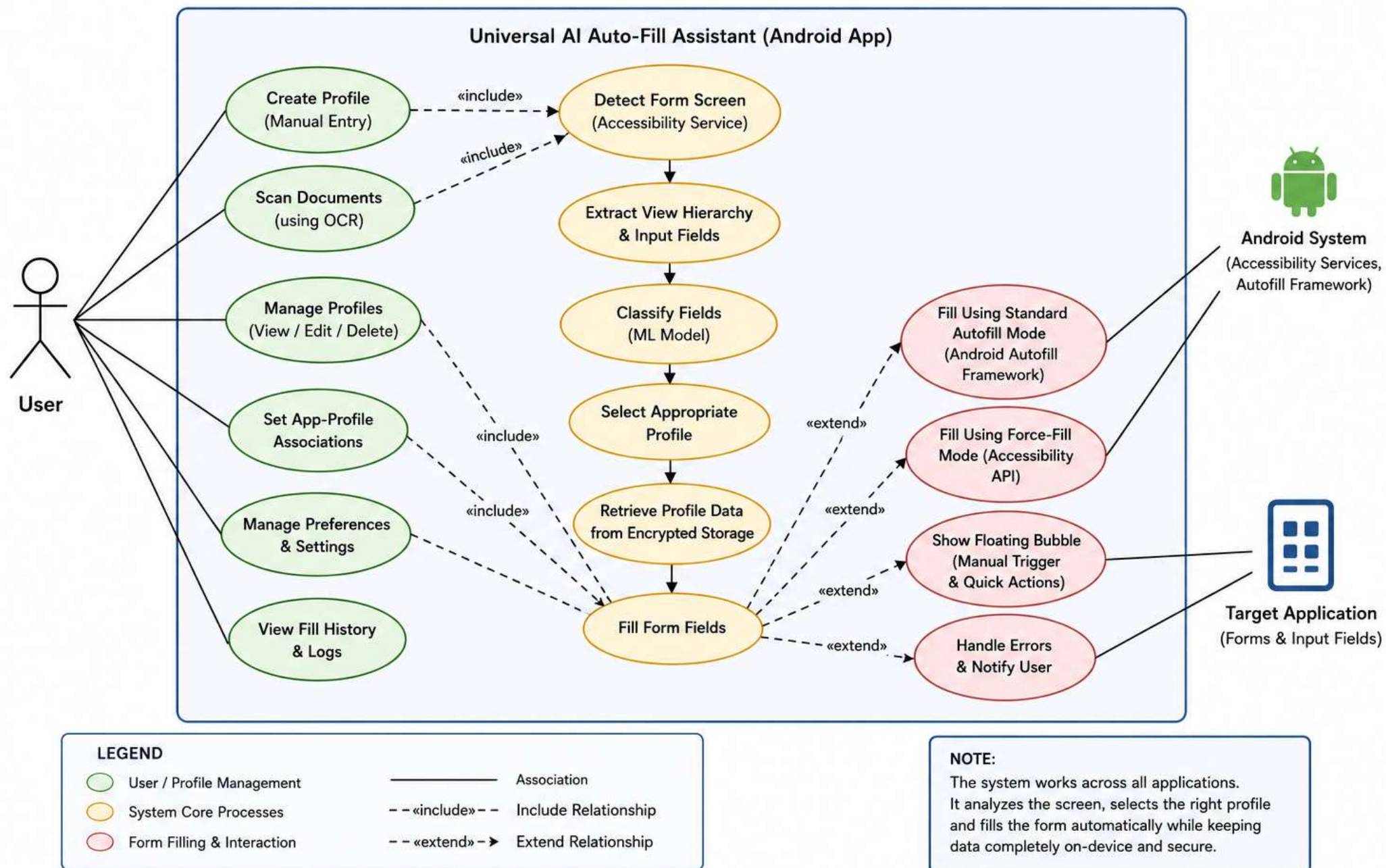
Target Application:

Contains forms and input fields where data is to be filled.

3. **Key Activities:**

- Create and manage user profiles .
- Scan documents using OCR .
- Detect form screens using accessibility services .
- Extract view hierarchy and input fields .
- Classify form fields using machine learning .
- Select appropriate user profile .

USE CASE DIAGRAM – UNIVERSAL AI AUTO-FILL ASSISTANT



✓1: Introduction to Class Diagram

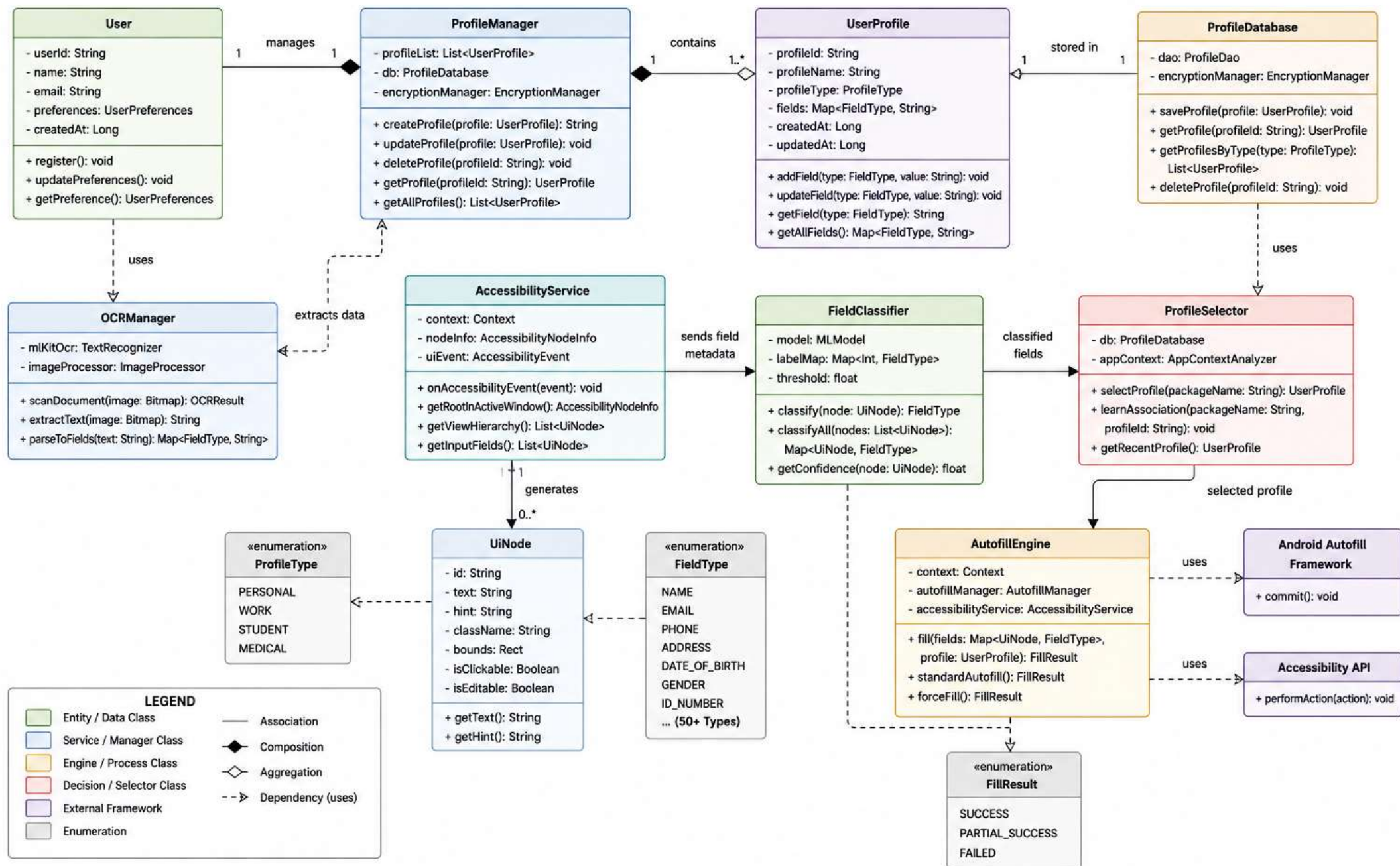
• Shows the **structure of the Smart AI Autofill System** and how different classes interact. Each class contains **attributes** (user data, profiles, settings) and **methods** (autofill, scan, detect, encrypt). Represents the **static design** of the Universal AI Autofill Assistant. Helps visualize data flow between modules before implementation. Defines responsibilities of each system component clearly.

✓2: Relationships Between Classes

Key Relationships:

- User → ProfileManager
- User → OCRScanner
- User → AutofillManager
- ProfileManager → UserProfile
- UserProfile → ProfileSection
- ProfileSection → SectionField
- OCRScanner → DocumentClassifier
- OCRScanner → UserProfile
- AccessibilityService → FieldDetector
- AccessibilityService → AutofillManager
- FieldDetector → MLFieldRecognizer
- AutofillManager → UserProfile
- DatabaseManager → UserProfile
- EncryptionManager → DatabaseManager
- SettingsManager → User

CLASS DIAGRAM – UNIVERSAL AI AUTO-FILL ASSISTANT



✓ 1: Introduction to Activity Diagram :-

Activity Diagram shows the **flow of actions** performed by the user and system.

Represents the **dynamic behavior** of the Smart AI Autofill System.

Explains the **step-by-step workflow**, decision points, and system responses.

Illustrates how user data is scanned, stored, detected, and auto-filled.

Helps understand interactions between user, AI modules, and form fields.

✓ 2: User Activity Flow

1. Open App

User launches the Smart AI Autofill application.

2. Create/Select Profile

User creates a new profile or selects an existing profile containing personal information.

3. Decision: Scan Document?

Yes → Open OCR Scanner and scan Aadhaar, PAN, Marksheet.

No → Manually enter profile information.

4. Store Profile Data

System saves user details in encrypted local storage.

5. Open Form

User opens any website or application containing a form.

6. Activate Autofill Bubble

User taps the floating AI Autofill bubble.

7. Detect Form Fields

System uses Accessibility Service and AI Field Recognition to detect visible form fields.

8. Match Profile Data

AI matches detected field labels with stored profile information.

9. Decision: Matching Data Found?

Yes → Autofill relevant fields automatically.

10. Fill Form

System inserts matched values into the detected form fields.

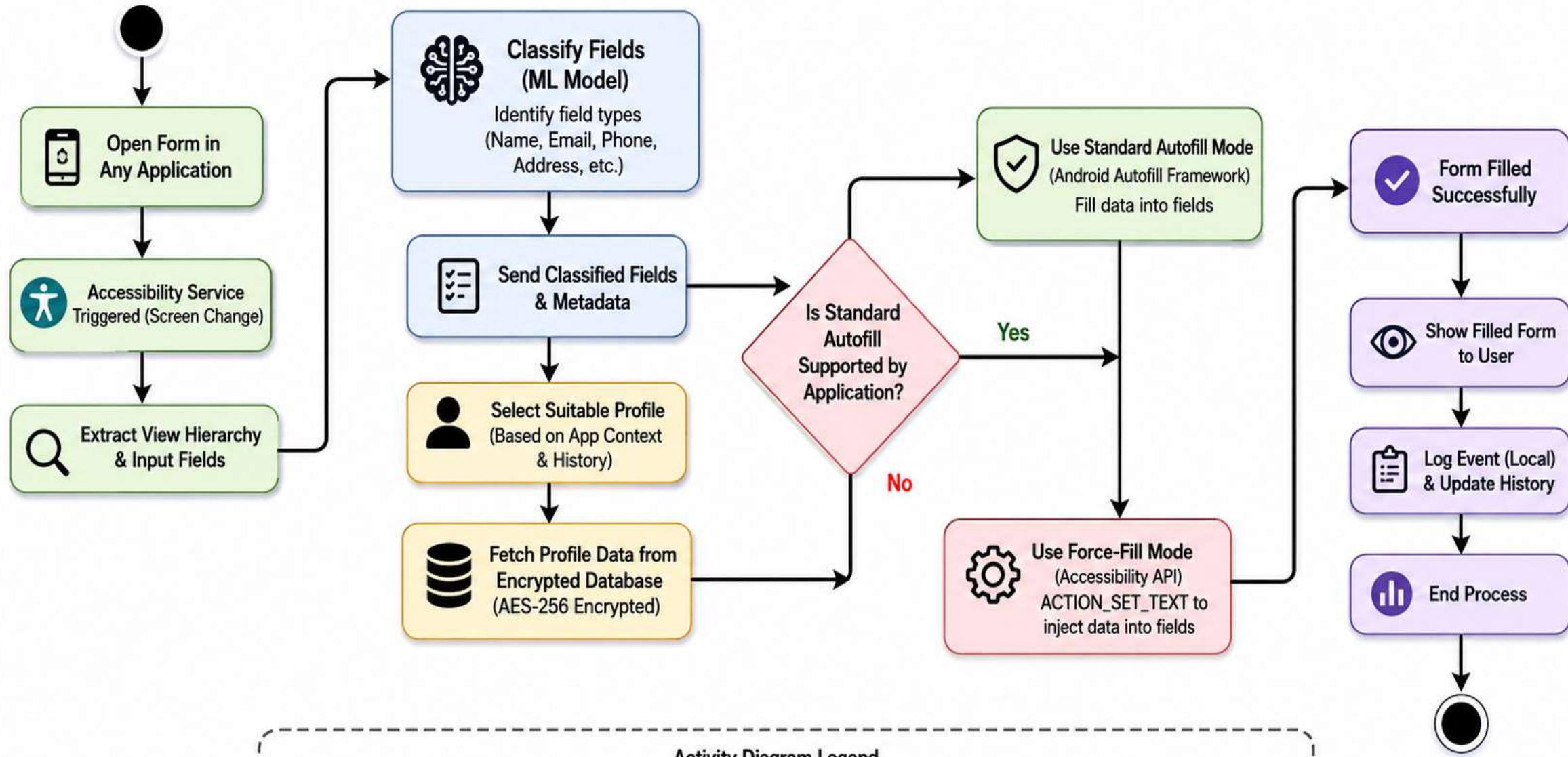
11. Review & Submit

User reviews the filled information and submits the form.

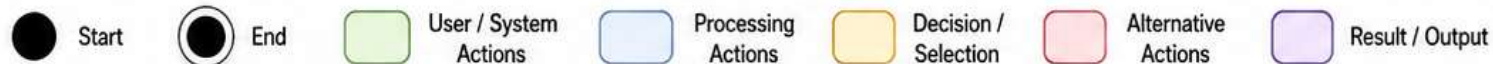
12. End Activity

Form filling process is completed successfully.

ACTIVITY DIAGRAM – UNIVERSAL AI AUTO-FILL ASSISTANT



Activity Diagram Legend



Sequence Diagram – Smart AI Autofiller

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Definition:

Shows interaction between **User, App, Accessibility Service, AI Field Detector, and Database** during autofill.

Flow Summary

1. App Launch

User opens the app.

Profiles and services are loaded.

2. Profile Selection

User selects/creates a profile.

App retrieves profile data.

3. Form Detection

User opens a form and taps the Autofill Bubble.

Accessibility Service detects form fields.

4. Autofill Process

AI identifies field types.

App fetches matching profile data.

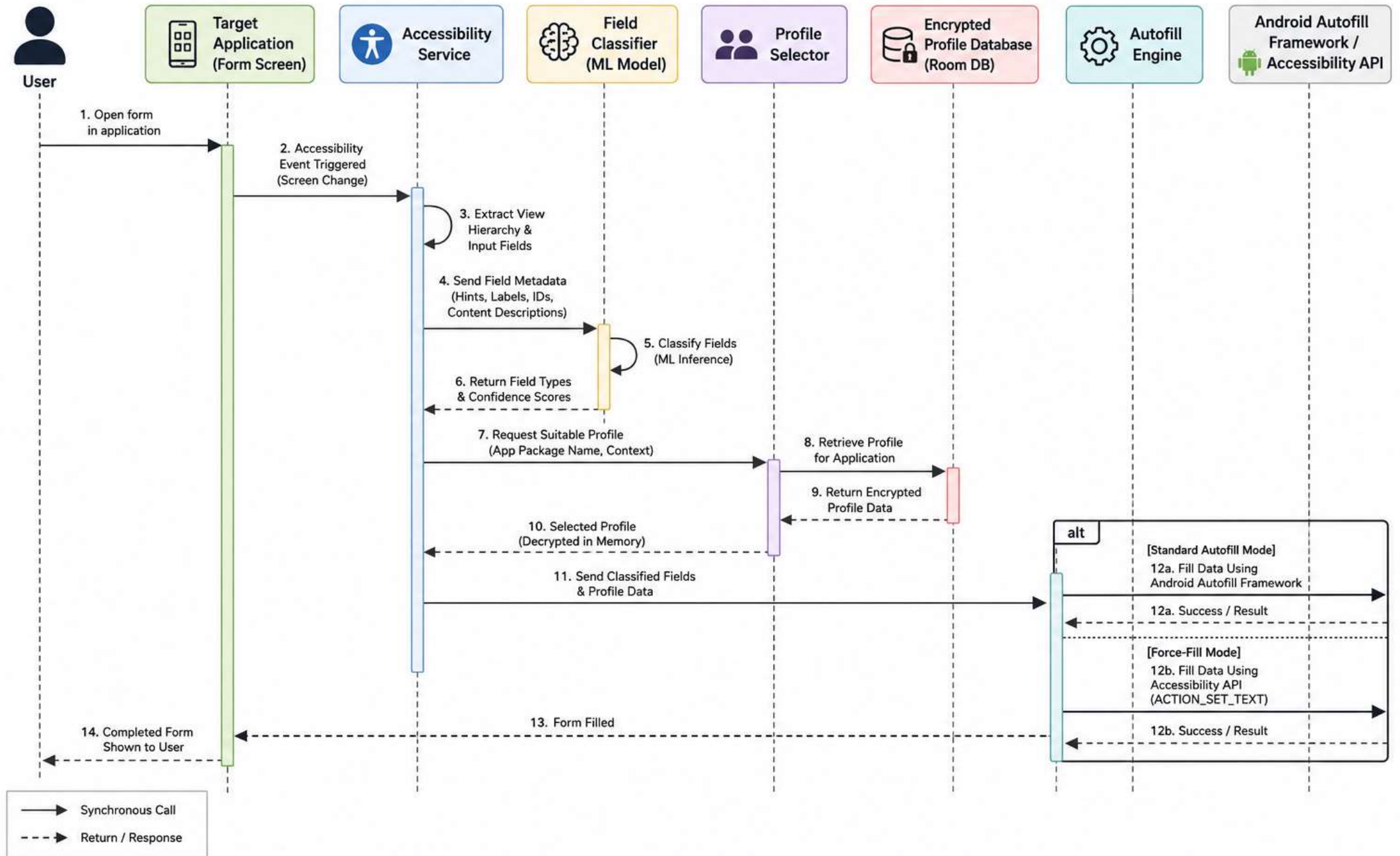
Form fields are filled automatically.

5. End Activity

User reviews and submits the form.

Autofill process completes.

SEQUENCE DIAGRAM – UNIVERSAL AI AUTO-FILL ASSISTANT



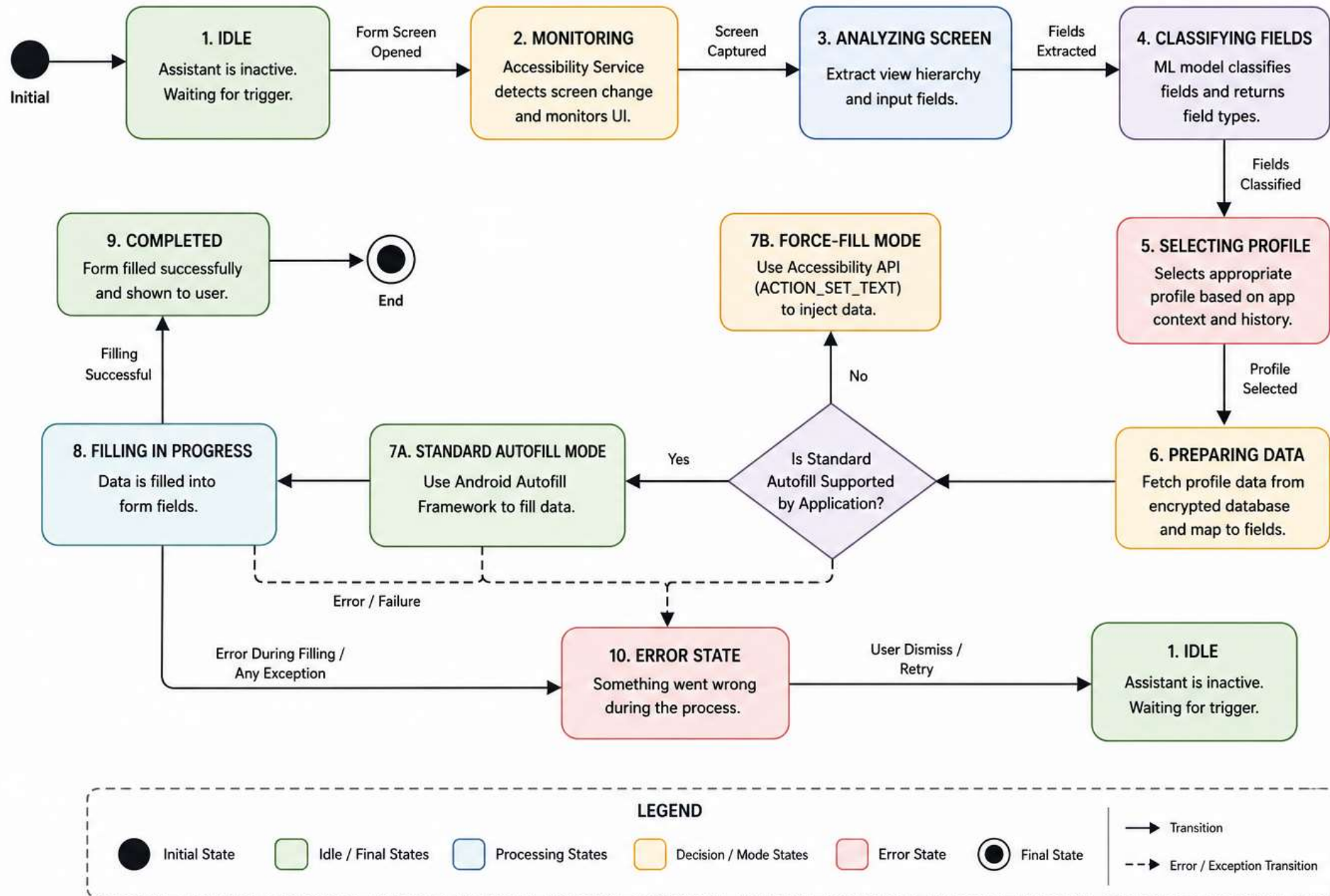
State Diagram – Smart AI Autofiller

Definition:

Shows how the app transitions between different states such as profile management, document scanning, form detection, autofill processing, and form submission.

- App launches → Initialize modules
- Load user profiles → Ready state
- User selects/creates profile → Profile Active
- User scans document → OCR extracts data
- Save data → Update profile
- User opens form → Detect form fields
- AI matches fields → Retrieve profile data
- Autofill process → Fill form automatically
- User reviews form → Submit or Edit
- Autofill completed → Ready for next form
- User exits → App Closed

STATECHART DIAGRAM – UNIVERSAL AI AUTO-FILL ASSISTANT



✓ 1: Introduction to Last Level DFD:-

- This is the detailed **Data Flow Diagram**, showing all major processes.
- It represents how data flows within the Smart AI Autofill System.

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1. Profile Initialization & Data Storage Flow

App Launched → Initialize Modules → Load User Profiles → Profile Available?

Yes → Retrieve Profile Data → Ready for Autofill

No → Create New Profile → Store Data in Database

2. Form Detection & Autofill Flow

User Opens Form → Activate Autofill Bubble → Detect Form Fields → AI Field Recognition → Matching Data Found?

Yes → Retrieve Data from Profile → Autofill Form Fields → Review Form

No → Request Manual Input → Update Profile Data

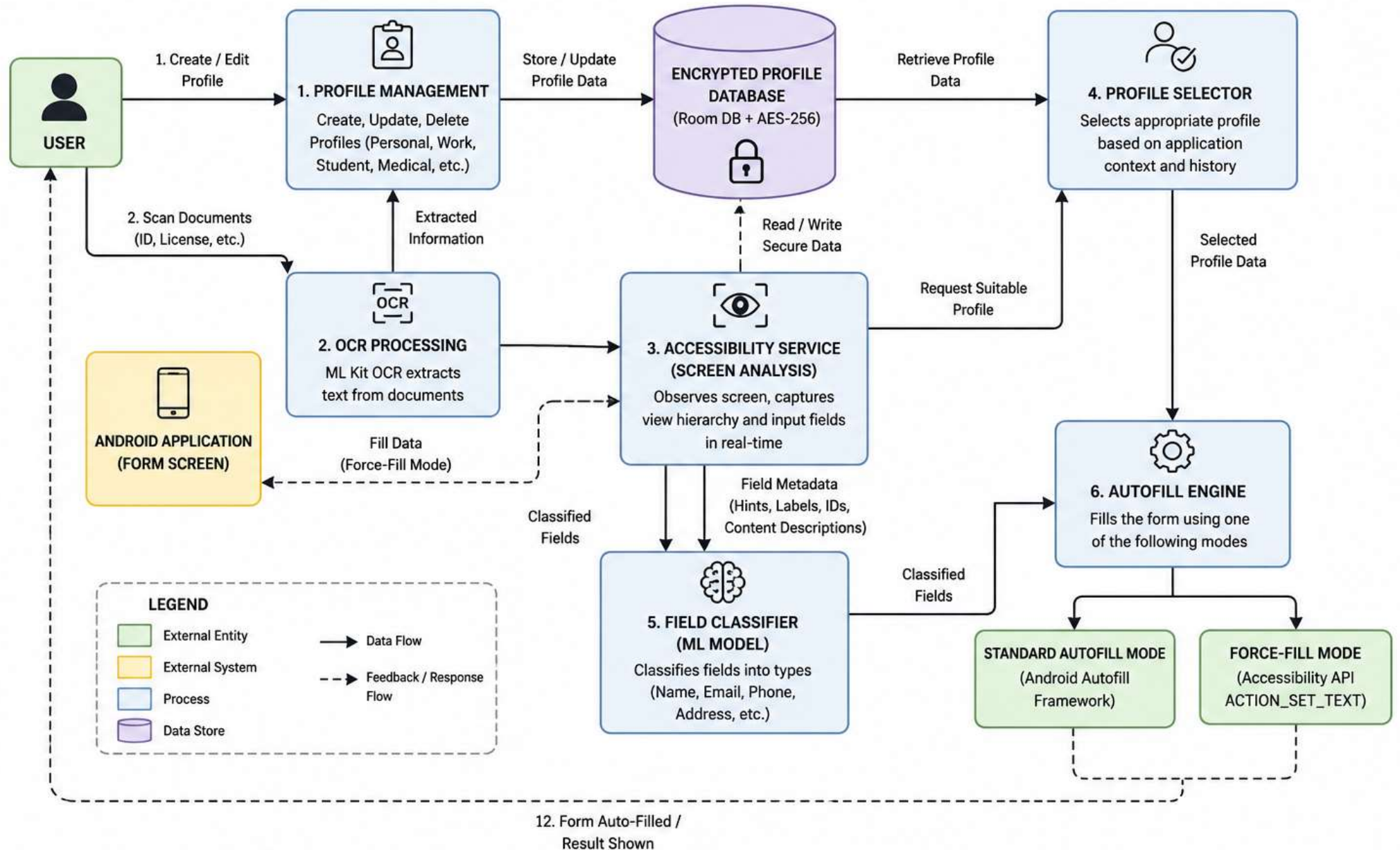
3. OCR Scan & Profile Update Flow

User Scans Document → OCR Extracts Information → Document Type Detected → Create/Update Profile Section → Save Data to Database → Profile Updated
(or OCR Failed → Retry Scan)

4. Form Submission & Completion Flow

User Reviews Form → Submit Form → Autofill Completed → Ready for Next Form → User Exit → App Closed
(or User Edits Data → Re-autofill Process)

DATA FLOW DIAGRAM – UNIVERSAL AI AUTO-FILL ASSISTANT





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Thank You

Queries ?